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A second-order meta-analysis on the effects of cooperative learning on students' academic achievement, higher-order thinking, and affective behaviors

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Abstract
This study aims to examine the impact of cooperative learning (CL) on learning outcomes, including academic achievement, high-order thinking skills, and affective behavior. In this regard, 15 first-order meta-analyses (FOM) studies were accessed using various inclusion criteria through Web of Science, Scopus, EBSCO, and Google Academic databases. The findings of FOM studies were synthesized through second order meta-analysis method. Statistical analyses were conducted under the random effects model. Heterogeneity and moderator analyses of the data set and examination of publication bias of effect sizes were performed. The results of this study reveal that CL has a moderate effect on learning outcomes (ES = 0.71 [CI = 0.55-0.87]). The impact of CL on learning outcomes was observed to statistically differ according to CL techniques, academic field, and the experimental design type of basic studies covered by FOM research.

Keywords Cooperative learning · Achievement · Higher order thinking · Affective behavior · Second order meta-analysis

Introduction

In order to equip students with the requisite knowledge to thrive in the knowledge society of the 21 st century, educational institutions are increasingly adopting social constructivist learning models. The knowledge society demands that individuals possess a range of higher-order cognitive skills, including problem-solving, creative thinking, critical thinking, and the capacity for creative thinking. It also requires developing social skills, such as the ability to work in teams, collaborate, lead, and communicate effectively. In this context, there is a growing interest in social constructivist learning models aimed at enhancing students' social and higher-order thinking skills. One such model is the cooperative learning model, which supports students in developing social skills, fostering positive attitudes towards learning and professions (Johnson & Johnson, 2009; Klang et al., 2020).

The practice of cooperative learning originated in higher education institutions in the United States during the 1970 s and has since become a pervasive phenomenon worldwide (Dyson et al., 2022; Johnson & Johnson, 2009). Its implementation has been observed across a diverse range of fields, spanning from elementary to higher education, including mathematics, social sciences, physical education, and vocational education programs (Baligar et al., 2022; Dyson et al., 2022). The relationship between cooperative learning and student relationships, social acceptance, and social inclusion has been examined by researchers, who have demonstrated a positive relationship with social acceptance (Klang

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et al., 2020). Moreover, comprehensive systematic evaluations have been conducted by researchers, synthesizing the findings of independent studies on CL (Dyson et al., 2022; Fernandez-Rio et al., 2022). In addition to these studies, first-order meta-analyses (FOMs) have been conducted to synthesize the statistical findings of independent studies on CL. However, FOMs have been conducted with limitations, including the restriction to a single country location (Stanczak et al., 2022; Tanjung et al., 2022) or a single learning outcome (Gürdoğan-Bayır & Bozkurt, 2018) or a single academic field, such as mathematics (Capar & Tarim, 2015). This situation increases the likelihood of product and location bias with regard to the efficacy of CL. It is imperative that comprehensive analyses of cooperative learning be conducted. It is, therefore, possible to draw general conclusions regarding the effectiveness of CL. Besides, by synthesizing evidence from multiple meta-analyses, a second-order meta-analysis reduces or minimizes second order sampling error that emerges in first-order meta-analyses (Schmidt & Oh, 2013). The objective of this research was to synthesize the findings of FOMs examining the impact of CL on learning outcomes. In other words, the objective of this research was to examine the impact of CL on learning outcomes based on the first-order meta-analysis studies. In this regard, the following questions were posed for investigation:

1. What is the level of effect of CL on students' learning outcomes?
2. Does the impact of CL on learning outcomes vary according to moderator variables?

Literature review

Cooperative learning

Cooperative learning is defined as "group learning activity organized so that learning is dependent on the socially structured exchange of information between learners in groups and in which each learner is held accountable for his or her own learning and is motivated to increase the learning of others" (Olsen & Kagan, 1992, p. 8). In a cooperative learning group, students of diverse abilities cooperate to reach shared objectives by engaging in discussions, helping each other grasp the material, and motivating one another (Slavin, 2014; Gillies & Ashman, 2003). Teachers play a key role by designing well-structured cooperative lessons, and regularly checking each student's contributions to ensure active involvement, which results in a collective improvement, with all members achieving greater academic success than they would alone (Johnson & Johnson, 1994). Therefore, teachers expert in CL are expected to consider five fundamental and crucial elements in their instruction: Positive interdependence, promotive face-to-face interaction, individual accountability, interpersonal and small group skills, and group processing (Dzemidzic Kristiansen et al., 2019; Johnson & Johnson, 1994, 1999, 2002).

One of the most significant aspects of cooperative learning is its positive reward interdependence, where students' success is mutually supportive (Slavin, 1980). Unlike competitive or independent structures, cooperative learning combines group rewards with individual accountability, consistently enhancing student achievement across various subjects and grade levels (Slavin, 1980, 1983). Furthermore, cooperative learning groups are kept small to facilitate meaningful face-to-face interactions and are composed of students with high-, medium-, and low- abilities. Students are taught and motivated to use essential social skills and engage in group processing, regularly discussing their progress and maintaining effective working relationships (Veenman et al., 2002).

Cooperative learning is an umbrella term that is the core of many active learning approaches (Kagan & Kagan, 2009), emphasizing teamwork, mutual support, and accountability (Johnson & Johnson, 2018). Problem-based learning, for example, relies on students working in small, cooperative groups to discover and master content collaboratively, building on cooperative learning principles. Similarly, team-based learning also focuses on mutual learning goals, reflecting the cooperative structure. Peer-assisted learning aligns with cooperative methods, emphasizing mutual encouragement rather than competition. Although cooperative learning and collaborative learning share more similarities than differences, a primary distinction lies in their approach to accountability and authority (Yang, 2023). While collaborative learning prioritizes autonomy and shifts authority to students by engaging them in open-ended tasks, cooperative learning emphasizes accountability through structured quantitative methods controlled by the teacher (Brufee, 1995).

Various instructional techniques have been developed for cooperative learning practices. Yang (2023) summarizes these techniques with their inventors as think-pair-share (Lyman), the jigsaw (Aronson), group investigation (Sharan & Sharan), jigsaw II (Slavin), student-team-achievement-division (Slavin), team-games-tournament (Slavin), team-accelerated instruction (Slavin), learning together (Johnson & Johnson), constructive controversy (Johnson & Johnson), three-step interview (Kagan), inside outside circle (Kagan), rally Robin (Kagan), numbered heads together (Kagan), and co-op co-op (Kagan). These cooperative learning techniques aim to enhance students' problem-solving, collaboration, creative thinking, teamwork, communication, and empathy skills (Ghaith & Awada, 2022). For instance,

specific studies have focused explicitly on the jigsaw technique, a widely utilized cooperative learning method. While Özkan and Çetinkaya-Uslusoy (2024) and Shakerian et al. (2020) documented the positive effect of the jigsaw learning technique on academic achievement, particularly in the medical education area, Stanczak et al. (2022) did not find a significant effect of jigsaw on the achievement of sixth-graders when compared to individualistic or teaching as usual methods.

Effect of cooperative learning on learning outcomes

The concept of learning outcomes is central to educational research, serving as the measurable change in knowledge, skills, or attitudes that results from instruction (Kirkpatrick & Kirkpatrick, 2006). Since the mid-twentieth century, the comprehensive classification of these outcomes has been guided by the hierarchical taxonomies of educational objectives. These taxonomies, primarily established by Bloom and his colleagues, define learning across three broad, fundamental domains: the Cognitive Domain (which focuses on mental skills and knowledge acquisition), the Affective Domain (concerned with attitudes, values, and interests), and the Psychomotor Domain (dealing with physical and manual skills) (Bloom et al., 1956; Krathwohl et al., 1964; Harrow, 1972). Adherence to this tripartite framework ensures that educational interventions, such as cooperative learning, are evaluated holistically across the entire spectrum of potential student development.

Previous meta-analyses on cooperative learning, published since 1988 across various academic fields in different countries, provide valuable insights into its impact on learning outcomes. While the theoretical framework for educational objectives encompasses the comprehensive cognitive, affective, and psychomotor domains (Bloom et al., 1956; Krathwohl et al., 1964; Harrow, 1972), the dependent variables (learning outcomes) included in the available first-order meta-analysis studies in the data set led to a data-driven operationalization of these domains. Specifically, the lack of suitable psychomotor-focused meta-analyses limited our scope to the cognitive and affective outcomes. Consequently, the dependent variables found in the included studies were categorized into the following three distinct groups: academic achievement (low-level cognitive), higher-order thinking skills (complex cognitive), and affective behaviors (affective domain).

Most meta-analysis studies have investigated the effect of cooperative learning on academic achievement, consistently demonstrating its positive effects across various educational contexts and disciplines. For instance, Çapar and Tarım (2015) analyzed 26 studies on the effect of cooperative learning on mathematics achievement, calculating a medium, positive, and statistically significant effect size. Similarly, Turgut and Gülşen Turgut (2018) reported moderate and positive effects of cooperative learning on mathematics achievement in Türkiye. Supporting the previous ones, Ridwan et al. (2022) confirmed the effectiveness of cooperative learning on mathematics learning outcomes in Indonesia with a medium effect size. Gürdoğan-Bayır and Bozkurt's (2018) meta-analysis also indicated a high, positive, and significant effect of cooperative approaches on social studies course achievement. Warfa (2016) shows a moderate positive effect of cooperative learning on chemistry achievement. Kyndt et al. (2013) considered 65 studies from 1995 to 2013 in primary, secondary, or tertiary education in face-to-face classroom environments and proved the positive effect of cooperative learning on achievement. Tanjung et al. (2022) proved that the investigation group-type cooperative model had an effect on physics achievement of 10th and 11th graders in Indonesia. Fuad et al. (2023) found that cooperative integrated reading and composition positively affected Indonesian students' reading literacy ability. Although these studies support cooperative learning's effectiveness, Setiana et al. (2020) found a lower effect size for cooperative learning's effect on mathematics achievement.

Some studies have included the effect of cooperative learning on affective factors which serves as the operational measure for the theoretical Affective Domain. Affective behaviors are specifically defined as the observable, measurable outcomes—such as reported attitude or performance confidence—that stem from underlying, theoretical affective factors like intrinsic motivation (Krathwohl et al., 1964). Based on the variables reported in the included first-order meta-analyses (e.g. Capar & Tarim, 2015; Kyndt et al., 2013; Liu & Lipowski, 2021; Ozkan & Çetinkaya-Uslusoy, 2024), this category specifically encompasses non-cognitive outcomes such as self-confidence, general learning perceptions, attitudes toward the subject, academic motivation, and intrinsic motivation. For instance, Tanjung et al. (2022) revealed very high effect of cooperative learning on affective competencies for K12 physics students in Indonesia. Liu and Lipowski (2021) highlighted the statistically significant effect of cooperative learning intervention on physical education students' intrinsic motivation. Kyndt et al. (2013) indicate a significantly positive effect of cooperative learning on attitude. For the jigsaw technique in cooperative learning, Özkan and Çetinkaya-Uslusoy (2024) revealed a positive effect on nursing students' attitudes. However, Çapar and Tarım (2015) reported a low effect size of cooperative learning on attitudes towards mathematics.

Higher-order thinking skills (HOTS) has gained central importance within contemporary pedagogical paradigms (Kim, 2025) and stands as another critical variable in cooperative learning research. HOTS are defined as complex

cognitive processes that are non-algorithmic, effortful, and involve self-regulation, often yielding multiple solutions or requiring nuanced judgment in the face of uncertainty (Resnick, 1987). Rooted in the cognitive domain of Bloom's Taxonomy (Bloom et al., 1956), these skills specifically correspond to the upper tiers of cognitive processing, encompassing analysis, evaluation, and creation. Within the literature, the term HOTS therefore operationalizes abilities such as critical thinking, inferential reasoning, and complex problem-solving (Ghanizadeh et al., 2020), all of which require the deep manipulation and restructuring of acquired knowledge. Siagian et al. (2023) underlined the positive effects of cooperative learning on enhancing mathematics students' critical and creative thinking. Also, the jigsaw technique seems to be effective variable in higher-order thinking skills. For instance, Solissa et al. (2023) found a statistically significant high effect of cooperative learning on students' higher-order thinking skills. Although Özkan and Çetinkaya-Uslusoy (2024) documented a positive effect of cooperative learning on critical thinking skills, non-significant p-values and high heterogeneity values might show the variance across studies.

Methods

This is a second-order meta-analysis study on the effects of cooperative learning on students' learning outcomes. Second-order meta-analysis- sometimes called as umbrella review, meta-meta-analysis or meta-analysis of meta-analyses (Cafri et al., 2010; Cooper & Koenka, 2012)- involves synthesizing a set of statistically independent and methodologically similar first-order meta-analyses that investigate the same underlying relationship across varying contexts (Schmidt & Oh, 2013). There are several first-order meta-analysis studies in the literature regarding the effect of CL on learning outcomes, thus providing a pool for a second-order meta-analysis; however, the first order meta-analysis studies have limitations and restrictions, culminating in the likelihood of product and location bias with regard to the efficacy of CL. To overcome these limitations and provide an overview of the literature, the statistical findings of first-order meta-analysis (FOM) studies examining the impact of CL on learning outcomes were synthesized using the second order meta-analysis method. Besides, first-order meta-analysis reduces sampling error by combining effect sizes from primary studies, but some error—known as second-order sampling error—remains due to the limited number of studies. Second-order meta-analysis aims to address this by synthesizing results from multiple meta-analyses to further refine and generalize effect size estimates (Schmidt & Oh, 2013). Additionally, this study was conducted following the PRISMA systematic review guidelines (Page et al., 2021).

Eligibility criteria

Various inclusion and exclusion criteria have been established in line with the objectives of this study. The inclusion and exclusion criteria are as follows:

i) FOM studies should focus on CL and learning outcomes. If the CL model is combined with a different small group learning model (e.g., collaborative learning, peer learning), it is excluded. ensure comprehensive coverage of educational outcomes, included studies must report on learning outcomes pertaining to the three established domains of educational objectives: cognitive, affective, and psychomotor. However, based on the results of the initial search and screening, the first-order meta-analyses available for inclusion predominantly focused on the cognitive and affective domains. Consequently, our final operational analysis is limited to the sub-categories derived from these two empirically represented domains. CL trials should involve face-to-face interactions. Studies on technology and web-supported CL are excluded.
ii) FOM studies must be articles published in English between 2010 and 2024. There were several reasons of defining this time span. The cooperative learning model has been carried out towards the end of the 2000 s by integrating digital technologies as well as traditional techniques. This transformation in educational technologies in 2000 s should be taken into consideration (Yang, 2023). Besides, systematic evaluation protocols such as PRISMA (2009) started to be used by researchers in 2010 (Page et al., 2021), and it can be accepted that this situation is reflected in the quality of the meta-analyses conducted. In addition to these situations, searches were made between 2010 and 2024 in order to access current evidence.
iii) FOM studies should encompass independent research conducted with experimental designs. In FOM studies, the traditional lecture-based method should be defined as the control group.
iv) FOM studies should contain sufficient statistical data to synthesize effect sizes (For example, Hedge's g, Cohen's d, Fisher's z, and Odds Ratio, and related statistics such as upper limit, lower limit, variance, and standard error.).
v) The overlap rate between FOM studies should be at most 25%. If the overlap rate is 25% or less, the FOM studies are considered independent from each other

(Cooper & Koenka, 2012). If the overlap rate exceeds 25%, the FOM study is excluded.

Search strategy

Web of Science, Scopus, and EBSCO databases were used to access FOM studies. The EBSCO database provides access to multiple databases. For this study, the ERIC, Academic Search Ultimate, MEDLINE Complete, Teacher Reference Center, and MasterFILE Complete Academic databases were selected. Additionally, the Google Academic database was used to access grey literature. Searches were conducted using binary combinations of keywords such as *cooperative learning*, *jigsaw*, *number head together*, *group investigation*, *student team achievement division*, *team games tournaments*, *cooperative integrated reading and composition*, *group discussion*, *think pair share*, *think talk write*, *creative controversy*, *investigation group*, *and meta-analysis*, *meta-analytic*, *systematic review*. For example, searches were conducted for combinations such as *cooperative learning and meta-analysis*, *cooperative learning and systematic review*, etc. Data flow is presented in Fig. 1. The characteristics of FOM studies included in this second-order meta-analysis are provided in Table 2.

Data extraction

The researchers created a coding form reflecting the characteristics of CL studies. FOM studies were independently coded by two researchers. The reliability coefficient Cohen's Kappa $\kappa = 0.89$ was calculated for the codings. Discrepant codes were discussed between the researchers, and in cases where no decision could be reached, education science experts were consulted. Codings were adjusted considering expert opinions. Coding procedure is presented in Table 1.

Quality assessment

The R-AMSTAR scale was used to assess the quality of FOM studies. The quality scores of the FOM studies included in the analysis ranged from 24 to 36. The study by Stanczak et al. (2022) is in the form of an internal meta-analysis. Therefore, it was decided that the R-AMSTAR scale is not suitable for assessing the quality of the Stanczak et al. (2022) study. Instead, the checklist developed by Tufanaru et al. (2017) for evaluating the quality of experimental research was used (Table 2).

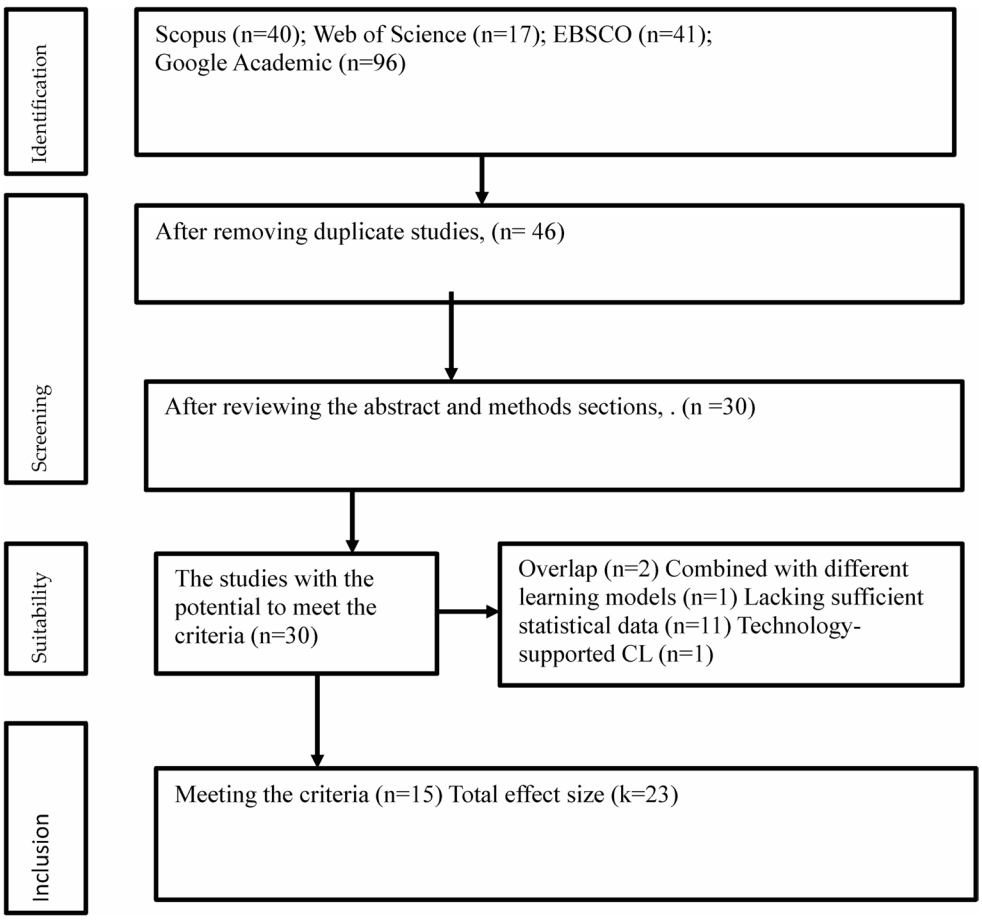


Fig. 1 Data flow diagram

Table 1 Coding procedure

Coding	
Group	Code
Study	Researchers (publication year)
Learning outcome	Learning outcomes were coded as academic achievement, high-order thinking skills, and affective behavior. If different learning outcomes of the primary studies were combined, they were coded as combined.
CL techniques	If different CL techniques were combined, they were coded as combined CL; if focused on only one CL technique, they were coded with the name of the technique. For example, jigsaw.
Academic scope	FOMs were coded according to the academic disciplines they covered. If multiple disciplines were evaluated together, they were coded as mixed.
Education level	The coding for the educational level covered by FOM studies includedas K12 and higher. If it covered multiple levels, it was coded as mixed.
Primary research design	FOM studies were coded as fully experimental, quasi-experimental, and fully or quasi-experimental based on the research designs they covered.
Primary research location	If the primary research comes from only one country, it was coded by country. If the primary research covers multiple countries, it was coded as "not specific location." Countries: Indonesia, Türkiye, other.
FOM quality	According to the score obtained from the Revised Assessment of Multiple Systematic Reviews (R-AMSTAR) scale (Kung et al., 2010), the ratings were coded as inadequate, low, medium, and high. No FOM was found at inadequate and low levels.
FOM publication bias status	If FOM detected publication bias, it was coded as "yes"; if it did not detect publication bias, it was coded as "no"; if publication bias was not reported, it was coded as "unclear".
FOM publication year range	between 2010 and 2024

The characteristics of FOM research in the dataset

The dataset for this study comprises 403 independent research studies. In this study, 15 FOM research studies were analyzed. These studies include FOMs from Indonesia ($n=5$), Türkiye ($n=4$), Switzerland ($n=1$), Iran ($n=1$), and various other countries ($n=4$). There are a total of one full experimental design studies, two quasi-experimental design studies, and 13 studies that encompass both full experimental and quasi-experimental basic research. Furthermore, eight of these studies are at the K12 education level, three at the university level, and four cover both K12 and university levels. Additionally, while four FOM studies focus on the jigsaw technique, one focuses on investigation group, and one on cooperative integrated reading technique, nine studies examine techniques related to the CL (Cooperative Learning) model. FOM studies have published effect sizes related to multiple learning outcomes (e.g., Ridwan et al., 2022; Kyndt et al., 2013). In this study, effect sizes that are independent of each other based on learning outcomes of FOMs have been coded. Therefore, multiple effect sizes are reported from some studies. In this study, k = 23 effect sizes were calculated. Liu and Lipowski (2021) reported odds ratios, while Setiana et al. (2020) reported Pearson correlation coefficients. Odd ratio and Pearson correlation coefficients were converted into Cohen's d indices (Borenstein et al., 2021). Additionally, effect sizes from other studies include Cohen's d (k = 7) and Hedge's g effect sizes (k = 14). Hedge's g value, which is preferred for small samples, is an adjusted version of Cohen's d value. For large samples, the values of these two indices are equal (Goulet-Pelletier & Cousineau, 2018; Marfo & Okyere, 2019). Therefore, Hedge's g and Cohen's d are combined in this study based on the assumption that the samples are large enough (Hew et al., 2021; Young, 2017). Thus, all effect sizes were converted into the same effect size index.

Synthesis method

In meta-analysis research, when the samples analyzed in studies are different from each other and their characteristics are also different, it is recommended to synthesize the findings under the random effects model (Borenstein et al., 2021). In this study, the effect sizes of FOM (first-order meta-analysis) research studies have been combined under the random effects model. The total heterogeneity of effect sizes and between-group heterogeneity Q statistics have been determined. Additionally, the I^2 statistic has been used to assess the level of heterogeneity of effect sizes, and tau^2 statistics have been used to interpret the source of heterogeneity. In this study, FOM research studies were grouped according to their characteristics, and moderator analysis was conducted for these groups. Furthermore, various publication bias tests have examined the distribution of effect sizes and whether they carry publication bias. Considering the advantages and disadvantages of publication bias tests, classic fail-safe test, funnel plot graph, Egger's test, and Duval & Tweedie Trim and Fill (DTTF) tests have been used in this study, and the results have been interpreted (Kepes et al., 2023).

Findings

In this study, k = 23 effect sizes were generated, ranging from ES = 0.00 to ES = 2.04. The mean effect size was calculated as ES = 0.71 [CI = 0.55–0.87.55.87]. The total heterogeneity of effect sizes was found to be Q(22) = 349.48, $p<.01$; with a heterogeneity level of I2 = 93.70 and $tau^2 = 0.13$. The classic fail-safe test for the distribution of effect sizes resulted in $N=4954$. Considering the critical value for the classic

Table 2 Characteristics of FOM studies in the data set

Study	ES	LL	UL	k	Scope	Level	Cooperative	Location	Design in FOM	Outcomes	Quality	Bias	Year range
Capar, & Tarim (2015)	0.59	0.38	0.80	36	Mathematics	Mixed	Cooperative	Other and Türkiye	Exp. and quasi-exp	Achievement	Medium	No	1988–2010
	0.52	−0.20	0.60	7	Mathematics	Mixed	Cooperative	Other and Türkiye	Exp. and quasi-exp	Affective	Medium	unclear	1988–2010
Fuad, et al. (2023)	2,04	1,35	2,74	10	Language	K12	Integrated reading	Indonesian	Quasi-experimental	Achievement	Medium	No	2015–2021
Gürdoğan-Bayır, & Bozkurt (2018)	0.96	0.72	1,19	32	Social Studies	K12	Cooperative	Türkiye	Exp. and quasi-exp	Achievement	Medium	unclear	2005–2016
Kyndt, et al. (2013)	0.54	0.47	0.60	51	Mixed	Mixed	Cooperative	Not specific	Exp. and quasi-exp	Achievement	High	Yes	1995–2012
	0.15	0.04	0.26	19	Mixed	Mixed	Cooperative	Not specific	Exp. and quasi-exp	Affective	High	Yes	1995–2012
	0.18	−0.05	0.40	4	Mixed	Mixed	Cooperative	Other and Türkiye	Exp. and quasi-exp	Affective	High	Yes	1995–2012
Liu, & Lipowskİ (2021)	0.05	−1,3	1,41	6	Physical education	K12	Cooperative	Not specific	Exp. and quasi-exp	Affective	Medium	Yes	2000–2020
Ozkan & Çetinkaya-Uslusoy (2024)	0.58	0.47	0.70	7	Nurse education	HE	Jigsaw	Not specific	Exp. and quasi-exp	Achievement	High	No	2011–2022
	0.70	0.13	1,26	4	Nurse education	HE	Jigsaw	Not specific	Exp. and quasi-exp	HOT	High	No	2011–2022
	1,18	0.53	1,84	6	Nurse education	HE	Jigsaw	Not specific	Exp. and quasi-exp	Affective	High	No	2011–2022
Ridwan et al., (2022)	0.87	0.63	1,10	10	Mathematics	K12	Cooperative	Indonesian	Quasi-experimental	Achievement	Medium	No	2013–2021
	0.92	0.70	1,14	11	Mathematics	K12	Cooperative	Indonesian	Quasi-experimental	Achievement	Medium	No	2013–2021
Setiana, et al. (2020)	0.30	0.08	0.56	16	Mathematics	K12	Cooperative	Indonesian	Exp. and quasi-exp	Achievement	Medium	No	2010–2020
Shakerian, et al. (2020)	1,25	1,08	1,51	7	Health Science	HE	Jigsaw	Iran	Exp. and quasi-exp	Combined	High	unclear	2010–2020
Siagian, et al. (2023)	0.79	0.58	1,00	35	Mathematics	K12	Cooperative	Indonesian	Exp. and quasi-exp	HOT	Medium	No	2012–2022
	0.69	0.42	0.98	24	Mathematics	K12	Cooperative	Indonesian	Exp. and quasi-exp	HOT	Medium	No	2012–2022
Solissa, et al.(2023)	0.84	0.66	1,03	10	Mixed	Mixed	Jigsaw	Not	Exp. and quasi-exp	HOT	Medium	No	2018–2023
Stanczak, et al. (2022)	0.00	−0.10	0.09	5	Mixed	K12	Jigsaw	Switzerland	Experimental	Achievement	High	unclear	Internal meta
Tanjung, et al. (2022)	0.96	0.74	1,17	15	Physics	K12	Investigation group	Indonesian	Exp. and quasi-exp	Achievement	Medium	unclear	2011–2020
	1,26	0.69	1,88	4	Physics	K12	Investigation group	Indonesian	Exp. and quasi-exp	Affective	Medium	unclear	2011–2020
Turgut & Gülşen Turgut.(2018)	0.84	0.68	0.99	59	Mathematics	Mixed	Cooperative	Türkiye	Exp. and quasi-exp	Achievement	Medium	unclear	2000–2017
Warfa (2016)	0.69	0.42	0.95	25	Chemistry	HE	Cooperative	Not specific	Exp. and quasi-exp	Achievement	High	Yes	2001–2015

exp. experimental, *HOT* Higher order thinking, *HE* Higher Education

fail-safe test as 125 (5k+10=5.23+10), it can be said that the calculation of the mean effect size is reliable. Egger's test did not indicate publication bias (t=2.08, p=.05). The DTTF test did not suggest any additional effect size that needs to be added. The distribution of effect sizes based on standard errors is presented in Funnel Plot Graph in Fig. 2. An examination of the funnel plot graph demonstratesthat the distribution of effect sizes is disrupted by the effect size located in the bottom left corner, affecting the symmetry. Considering the above publication bias tests, it was concluded that the effect sizes did not exhibit publication bias.

Considering the analyses of publication bias, total heterogeneity, and the level of heterogeneity, the effect of Cooperative Learning (CL) on learning outcomes is moderate. Furthermore, the effect sizes exhibit a high level of heterogeneity, and a significant source of this heterogeneity stems from differences between studies.

The between-group heterogeneity of effect sizes and effect sizes according to moderator variables are presented in Table 3. Remarkable findings from Table 3 are as follows.

The effect sizes vary statistically according to CL techniques (Q(3)=9.36, p=.02). While jigsaw techniques produce a moderate effect size (ES=0.72 [CI=0.40–1.04.40.04]), investigation group yields an effect size of ES=1.08 [CI=0.50–1.66.50.66], and cooperative integrated reading results in high effect sizes (ES=2.04 [CI=1.05–3.03]).

The impact of CL on learning outcomes varies statistically across academic domains (Q (6)=19.47, p<.01). In the Language domain, CL has a significantly high impact on learning outcomes (ES=2.04 [CI=1.05–3.03]), while in Physical Education, no significant effect was found (ES=0.05 [CI=−1.44−1.54]). High effects were observed in Science and Social Studies domains, while a moderate effect was observed in Mathematics.

Effect sizes also vary statistically based on the designs of primary research (Q(2)=9.83, p=.01). FOM studies encompassing full experimental designs did not yield any effect. Studies with quasi-experimental designs produced high effect sizes (ES=1.11 [CI=0.72−1.50]), whereas FOMs covering both full experimental and quasi-experimental designs resulted in moderate effect sizes (ES=0.69 [CI=0.54–0.84.54.84]).

Discussion

This study employed a second-order meta-analysis approach to integrate the effect sizes of 15 FOM research studies, with the objective of examining the impact of CL on certain learning outcomes. The distribution of effect sizes was analysed in accordance with the characteristics of FOM research studies, including heterogeneity and moderator analyses. The findings of this study indicate that CL exerts a moderate influence on learning outcomes. In other words, CL exerts

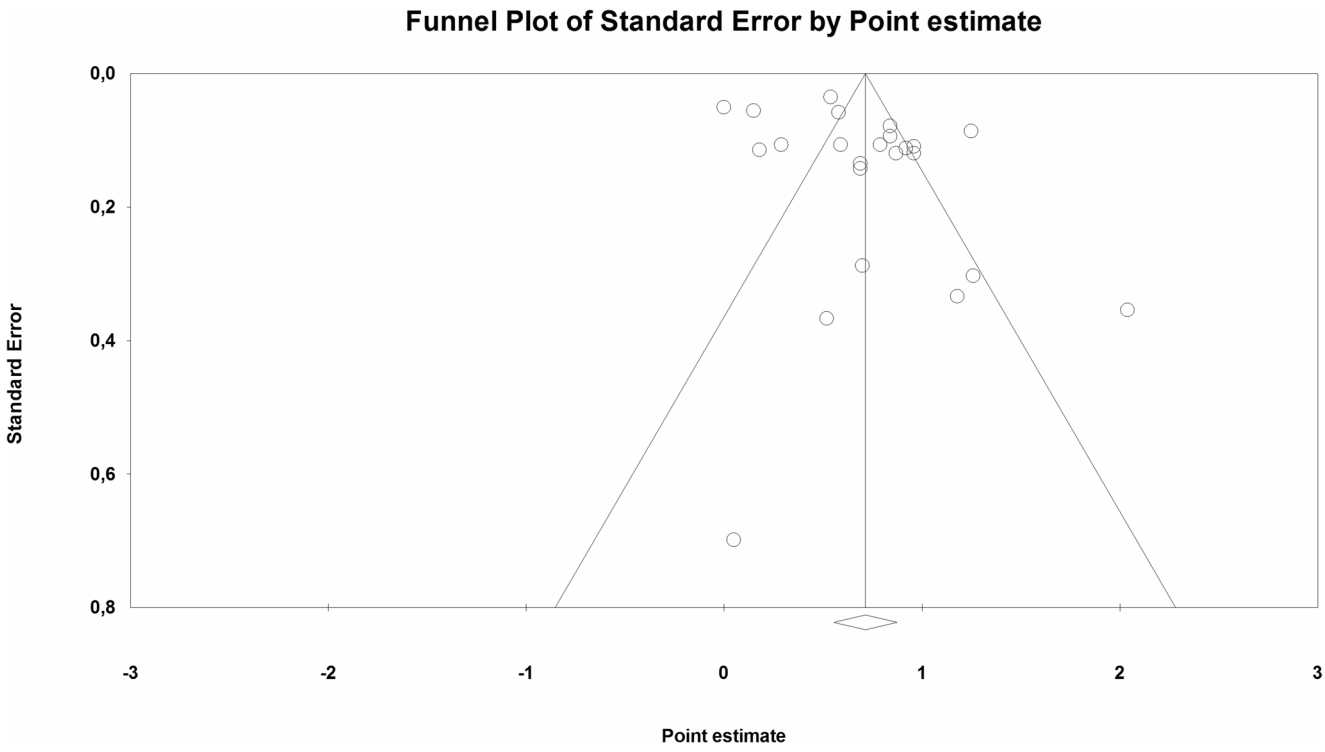


Fig. 2 The funnel plot graph

Table 3 Moderator and heterogeneity analysis

Group	k	ES	LL	UL	Q(b)	df	p
Outcomes					4.35	3	.23
Achievement	12	.71	.52	.90			
Affective	6	.51	.17	.84			
Higher order thinking	4	.76	.42	1.11			
Combined	1	1.25	.62	1.88			
CL tech					9.36	3	.02
Cooperative	14	.61	.40	.81			
Jigsaw	6	.72	.40	1.04			
Investigation group model	2	1.08	.50	1.66			
Cooperative integrated reading	1	2.04	1.05	3.03			
Academic scope					20.16	5	<.01
Mathematics	8	.70	.47	.93			
Science	7	.92	.66	1,17			
Language	1	2.04	1.13	2.95			
Social Studies	1	.96	.33	1.59			
Physical education	1	.05	−1.44	1.54			
Mixed	5	.34	.07	.61			
Education level					2.56	2	.28
K12	11	.79	.54	1.04			
Higher	5	.87	.50	1.24			
Mixed	7	.52	.23	.82			
Primary research location					2.55	3	.47
Indonesian	8	.90	.62	1.18			
Türkiye	5	.63	.27	.98			
Not	8	.61	.32	.90			
Other	2	.61	.09	1.14			
Primary research design					9,83	2	.01
Experimental and quasi-experimental	19	.69	.54	.84			
Quasi-experimental	3	1.11	.72	1.50			
Experimental	1	.00	-.58	.58			
First order meta analysis quality					3.69	1	.05
High	9	.54	.31	.77			
Medium	14	.83	.64	1,02			
FOM publication bias status					4.23	2	.12
No	11	.80	.56	1,05			
Yes	5	.37	-.01	.74			
Unclear	7	.82	.51	1,13			
FOM publication year range					1.51	1	.22
Before 2020	10	.60	.36	.85			
After 2021	13	.82	.58	1.05			

a moderate influence on students' academic achievement, higher-order thinking, and affective behaviours. The majority of previous studies also corroborate the moderate effect of CL on learning outcomes (e.g., Çapar & Tarım, 2015; Ridwan et al., 2022; Turgut & Gülşen, 2018). However, there is a paucity of studies that have demonstrated a high effect of CL (e.g., Gürdoğan-Bayır & Bozkurt, 2018) or a low effect (Setiana et al., 2020).

The analysis of the learning outcomes revealed that, although the difference among outcome types was not significant, CL demonstrated the highest effect size for higher-order thinking skills, followed by academic achievement. The considerable effect size indicates the potential of CL for enhancing higher-order thinking skills. In light of the impact of CL on critical and creative thinking skills (Huang et al., 2017; Silva et al., 2022), it would be prudent to subject CL to further scrutiny in order to ascertain its potential in this promising area. The effect size for affective outcomes is relatively smaller. Furthermore, some studies reported combined results (i.e. the combination of effect sizes for academic achievement and critical thinking). It is notable that the highest effect sizes were observed for combined outcomes. The findings of this study are consistent with previous research and indicate that CL is an effective approach

for fostering the development of higher-order thinking skills and academic achievement in students. Specifically, a number of studies have similarly indicated that the implementation of CL not only has a positive effect on academic achievement but also on higher-order thinking skills (Lu et al., 2021; Marcos et al., 2020; Takko et al., 2020).

The impact of CL on learning outcomes showed variation across academic domains, with the language domain exhibiting the largest effect size, followed by social studies, science, and mathematics. This finding appears to contradict earlier meta-analyses, such as that by Kyndt et al. (2013), which reported higher effect sizes in the hard sciences (e.g., science and mathematics) compared to the soft sciences (e.g., social sciences and languages). Their suggestion was that this variance might stem from the efficacy of student collaboration in tasks requiring structural hierarchy, common in subjects like mathematics (Lou et al., 1996). Crucially, the higher effect sizes observed for language and social studies in the present study must be interpreted with caution, as this second-order meta-analysis reveals a significant data limitation: these domains are each represented by only one First-Order Meta-Analysis (FOM) study in our dataset. The unique value of employing a second-order meta-analysis here is not just to report a finding, but to highlight this exact methodological gap: there is a severe paucity of high-level synthesis (meta-analyses) that systematically report effect sizes for the language or social studies domains. The under-representation of studies in these domains underscores a critical imbalance in the current research synthesis landscape, which severely limits the ability of the field to interpret the true impact of CL in soft sciences. While the findings related to specific academic domains are necessarily tempered by the low representation of soft sciences (i.e., a single FOM for language and social studies), the second-order meta-analysis methodology employed here proves its singular value by unmasking this precise research gap. Our study not only confirms the overall efficacy of CL but, more critically, establishes a clear agenda for future first-order meta-analyses, which must actively seek out and synthesize primary research in disciplines such as language and social studies to rectify this systemic imbalance in the synthesis landscape.

The impact of CL is not statistically significant in the context of physical education, when compared to other academic disciplines such as languages, social studies, science, and mathematics. However, this finding conflicts with previous studies (e.g., Bores-García et al., 2021; Zach et al., 2023) which have demonstrated a positive impact of CL on cognitive and affective learning outcomes in physical education. This discrepancy may be attributed to differences between the objectives of CL and the learning goals of academic domains. Furthermore, it is possible that teachers of physical education may be inclined to utilise conventional methodologies that prioritise physiological activation over social and educational components (Hortigüela-Alcalá et al., 2020). Yet, it should be noted that there is only one meta-analysis study regarding physical education in the data-set of the current study. The primary objective of cooperative learning is to cultivate students' social skills, including the capacity for collaboration, communication, and leadership. Nevertheless, an analysis of the distribution of FOM studies according to academic domains reveals a disproportionate focus on STEM fields, with a relative paucity of studies in language and social sciences. It is important to note that the effect sizes vary considerably in these fields. In light of the aforementioned purpose of cooperative learning, it is imperative to assess the efficacy of cooperative learning in the domains of social sciences, language and physical education. It is noteworthy that the extant literature is inadequate in this regard. In light of the above, we propose that further investigation be conducted into the effect of CL across academic domains.

Significant differences were observed in the effect sizes associated with the various CL techniques. The findings of this study indicate that the Investigation Group and Cooperative Integrated Reading and Composition (CIRC) CL techniques show a more pronounced impact on learning outcomes compared to others. This observation aligns with existing research that has highlighted the existence of differences in effectiveness between various CL techniques; for example, studies have found that the CL techniques of Student Teams' Achievement Division (STAD) and Team-Assisted Individualisation (TAI) (Turgut & Gülşen, 2018), as well as STAD and Learning Together (Çapar & Tarim, 2015), generally produce higher effect sizes. These collective findings lend support to the proposition that learning outcomes vary significantly according to the specific CL techniques employed. However, it is a necessary limitation of this study that the primary data coding was based on research outcomes rather than a systematic classification of CL techniques across all studies. Therefore, while our data suggest variations in effect sizes based on technique, it would be premature to conclude definitive superiority of one technique over another based solely on these results. Future research should prioritize coding first-order meta-analysis (FOM) studies systematically based on the specific CL techniques used to gain a deeper, more robust understanding of their differential impact on learning outcomes.

Furthermore, the effect sizes differed depending on the design types of primary research covered by FOM studies. A full experimental design was not found to yield any effect in a FOM study; however, studies employing quasi-experimental designs demonstrated moderate effect sizes. The number of effect sizes from studies encompassing both

full experimental and quasi-experimental designs provides substantial evidence of moderate effects. The notable effect sizes observed in quasi-experimental designs may be attributed to the fact that the interventions in question reflect real-world educational practices rather than the controlled conditions typically found in laboratory settings (Rogers & Révész, 2019). One noteworthy finding was the observation of a higher effect size in medium-quality meta-analyses when compared to high-quality ones, although this difference was not statistically significant. This raises the question of whether high-quality meta-analyses result in more conservative effect sizes due to their stricter inclusion criteria.

In addition to the learning outcome types, the study did not identify a statistically significant difference between the moderator variables of education level, primary research location, FOM quality, publication bias status, and FOM publication year range. Although the observed effect sizes did not differ significantly, the effect size for studies conducted at the post-secondary level was higher than for studies conducted at the K-12 level. The results pertaining to the age groups were inconclusive in the preceding meta-analyses. While Lou et al. (1996) and Kyndt et al. (2013) report significant effect sizes in favor of the primary level (6–12 years) and the tertiary level (18 years and older) and the late elementary level (9–12 years), Qin et al. (1995) found no significant differences for the age groups. The considerable effect size observed in the Indonesian context may also serve to substantiate the assertion put forth by Butera and Buchs (2019) that Western societies do not ascribe a high value to the social and cultural aspects of cooperation. The elevated effect sizes observed in the first-order meta-analysis studies published subsequent to 2021 may be indicative of an increasing familiarity among students with the use of cooperative and collaborative learning as contemporary pedagogical methods.

Limitations and further research

The conclusions of this second-order meta-analysis should be considered in light of several methodological and data-related constraints imposed by the existing research synthesis literature. A primary limitation arises from constraints in data categorization and classification. Although the theoretical framework for educational objectives encompasses the cognitive, affective, and psychomotor domains, the data categorization was driven solely by the variables present in the available first-order meta-analyses (FOMs), resulting in the necessary exclusion of the psychomotor domain due to insufficient empirical representation. Furthermore, our analysis of specific cooperative learning techniques (e.g., Investigation Group, CIRC) was limited because the original FOM studies primarily coded effect sizes by learning outcome rather than systematically by instructional technique, preventing us from concluding definitive superiority among methods. Similarly, the length of CL implementation was not included as a moderator variable since this crucial data was absent from the included FOMs; therefore, further synthesis studies should prioritize analyzing the effect of the implementation period length when such data becomes available.

Additional limitations pertain to sample size and geographic context. The analysis of academic domains was hindered by an insufficient sample size, as several soft science fields (e.g., language, social studies, and physical education) were each represented by only a single or very few FOM studies. This constraint necessitates caution when interpreting domain-specific findings and highlights a critical imbalance in the existing research synthesis landscape. Furthermore, the dataset is primarily drawn from research conducted in English and heavily features studies originating from Indonesia and Türkiye. This concentration increases the probability of location and language bias in the calculated effect sizes, suggesting that a future avenue for investigation would be multi-lingual meta-analyses covering a wider range of countries. Finally, this study is confined to the analysis of published articles, and a similar research project could be conducted to encompass doctoral and master's theses to potentially reduce publication bias.

Regarding the scope of the model, the present study is confined to an examination of the cooperative learning model and the dynamics of small-group interactions. Further synthesis studies could be conducted to examine related constructivist small-group interaction-based models, including collaborative learning, peer learning, and team-based learning. Lastly, the study incorporates the analysis of affective behavioral types collectively, but a more granular meta-analysis examining the impact of CL on specific affective behavior types (e.g., intrinsic motivation vs. self-confidence) could be conducted to access more detailed information.

Conclusion and recommendations

The traditional transmission approach to teaching and learning is inadequate to meet the educational demands of equipping students with 21st-century skills. Cooperative learning may be employed as a means of addressing these skills (Keramati & Gilles, 2024). This is why a variety of cooperative learning techniques are being increasingly employed in educational settings, and a number of studies have sought to combine the findings of individual studies (Capar & Tarim, 2015; Stanczak et al., 2022; Tanjung et al., 2022). However, no previous studies have yet attempted to combine the results of the FOM studies on cooperative learning.

The present study employed a second-order meta-analysis to combine the findings of the FOM studies, reporting that CL has a moderate effect on students' learning outcomes, with varying degrees observed among academic achievement, affective and higher-order thinking skills.

The particular findings of this study are of interest to those engaged in the field of education who are considering the use of CL in their teaching practice. The findings of this study demonstrate the efficacy of CL in improving student outcomes. Consequently, it is recommended that this learning model be employed to a greater extent in educational settings. It is recommended that in-service training courses be organized with the objective of enhancing the competencies of teachers and instructors in the techniques of CL. The application of CL in the classroom requires a comprehensive understanding of its theoretical underpinnings and practical implementation (Hortigüela-Alcalá et al., 2020). It is imperative that teachers are equipped with the requisite knowledge, skills and competencies. In order to facilitate learning, it is essential to provide students with sufficient time and structured experiences (Fernandez-Rio et al., 2022; Legrain et al., 2019). Despite the fact that teachers believe they are utilising CL, the activities in question may be too disparate and not aligned with a specific methodology (Morgan, 2019). Furthermore, web-based platforms can be employed to disseminate exemplars of CL technique applications among teachers and instructors. The study revealed that specific CL techniques yielded higher effect sizes. It is incumbent upon teachers to be knowledgeable about the various types of CL techniques that are most efficacious for their target student population and domain of study.

Author contributions FG conceived the idea for the second-order meta-analysis and designed the study framework in collaboration with MK and CE. İA led the systematic literature search and the selection of primary meta-analyses, with input from ATA. Screening and data extraction were performed by MK, with cross-verification by İA. MK developed the statistical methodology for aggregating effect sizes and analyzing heterogeneity across primary meta-analyses. MK conducted the statistical analyses, with oversight by CE. FG prepared the initial manuscript draft, including the introduction, methods, results, and discussion sections. All authors contributed to the critical review and revision of the manuscript for important intellectual content. FG supervised the overall project and coordinated the contributions of all authors. All authors approved the final manuscript and are accountable for the work presented in this study.

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Declarations

Ethical statements This second-order meta-analysis is based on publicly available data from previously published meta-analyses. As such, no new data were collected directly from participants, and no ethical approval was required. The study adheres to ethical standards for research, including accurate reporting, transparency, and proper attribution of all sources.

Consent to publish All authors affirm that they have read and approved the final manuscript and confirm their consent to publish. The authors guarantee that the manuscript does not contain any images, videos, or personal data from specific individuals that would require individual written consent for publication.

Consent to participate This study constitutes a second-order meta-analysis based entirely on the synthesis of previously published, aggregated research data (secondary sources). The research did not involve human participants, interventions, or the collection of new primary data. Therefore, the requirement for individual participant informed consent is not applicable for this study.

Conflict of interest There is no conflict interest in this study.

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